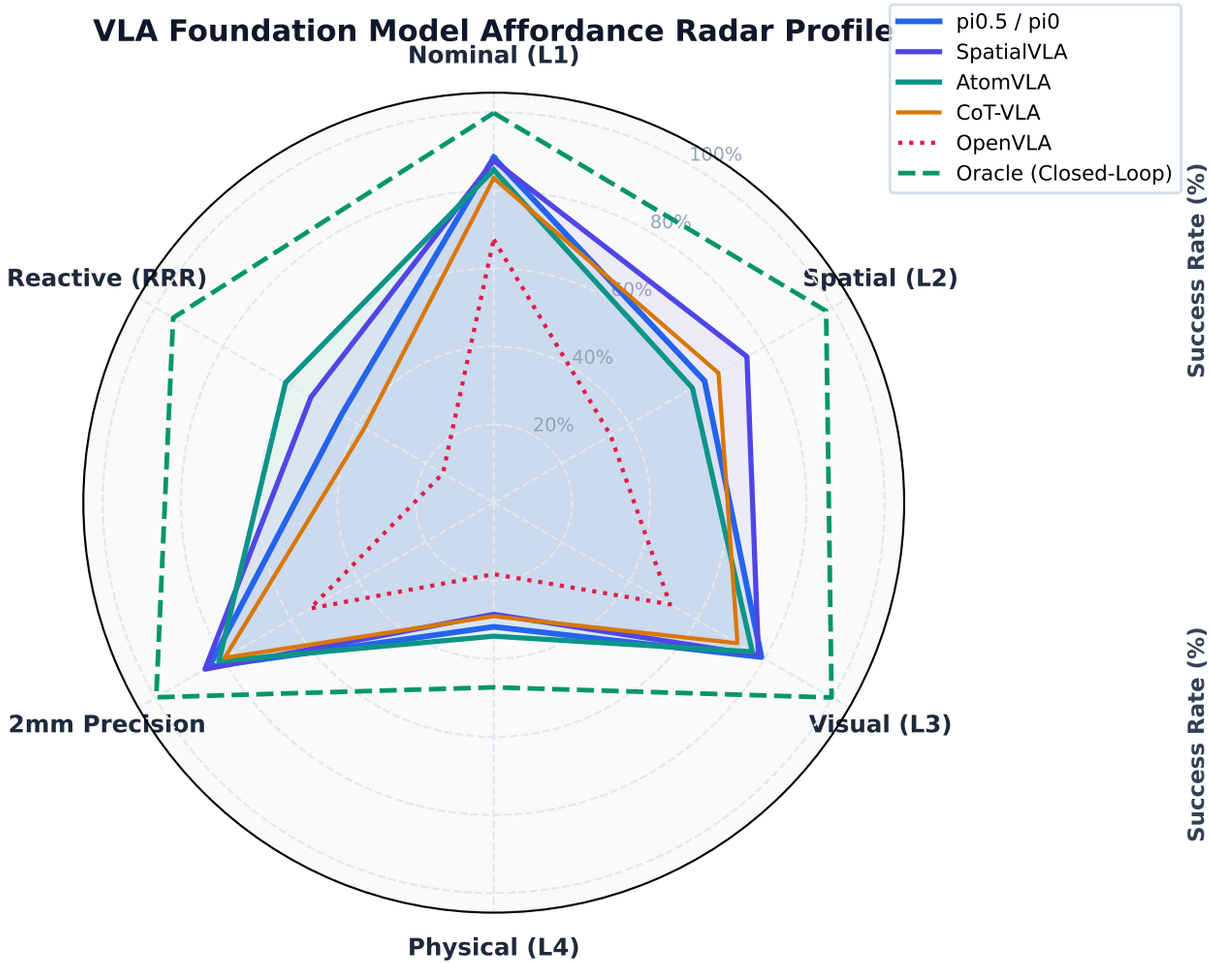
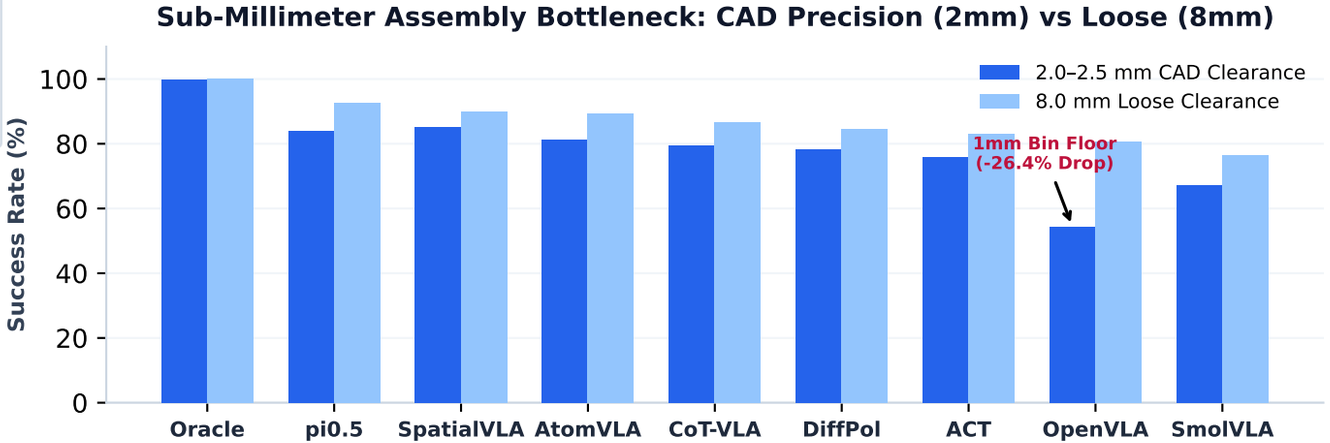
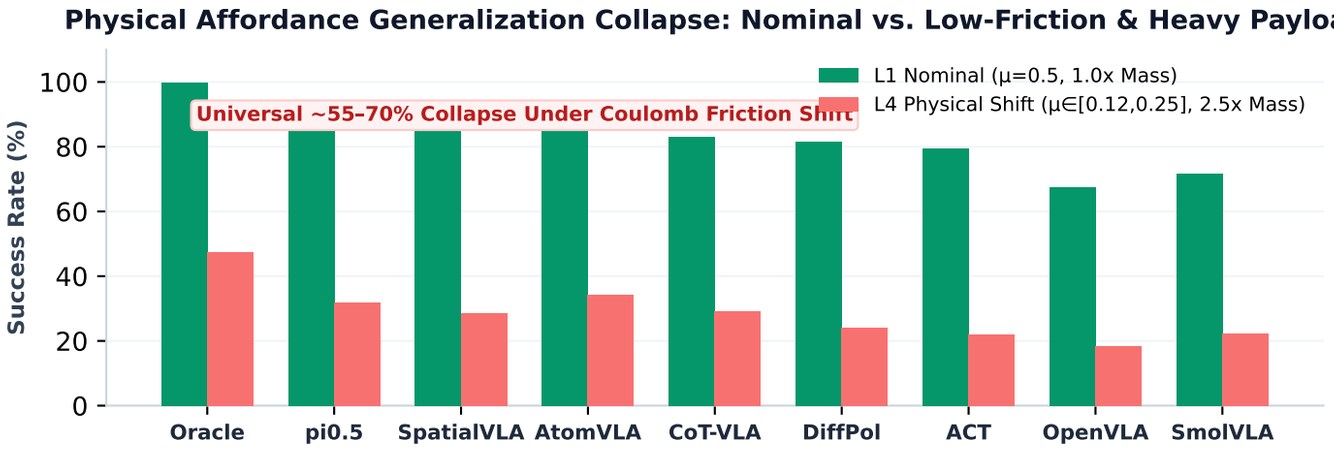

VLA Foundation Model Affordance Radar Profile
pi0.5 / pi0
SpatialVLA
AtomVLA
CoT-VLA
OpenVLA
Oracle (Closed-Loop)
Nominal (L1)
Spatial (L2)
Visual (L3)
Physical (L4)
2mm Precision
Reactive (RRR)
20%
40%
60%
80%
100%


Sub-Millimeter Assembly Bottleneck: CAD Precision (2mm) vs Loose (8mm)
Success Rate (%)
2.0–2.5 mm CAD Clearance
8.0 mm Loose Clearance
1mm Bin Floor
(-26.4% Drop)
Oracle
pi0.5
SpatialVLA
AtomVLA
CoT-VLA
DiffPol
ACT
OpenVLA
SmolVLA


Physical Affordance Generalization Collapse: Nominal vs. Low-Friction & Heavy Payload
Success Rate (%)
L1 Nominal (μ=0.5, 1.0x Mass)
L4 Physical Shift (μ∈[0.12,0.25], 2.5x Mass)
Universal ~55–70% Collapse Under Coulomb Friction Shift
Oracle
pi0.5
SpatialVLA
AtomVLA
CoT-VLA
DiffPol
ACT
OpenVLA
SmolVLA